MTH303 - Midsem 2

1. (3 points) Show that $\{\forall x(\alpha \to \beta), \forall x\alpha\} \vDash \forall x\beta$. (Context: $\vDash$ here is logical implication)

2. (4 points) State the compactness Theorem. Use it to prove that if $\Sigma \vDash \beta$, then there is a finite subset $\Sigma_0 \subseteq \Sigma$ such that $\Sigma_0 \vDash \beta$. (Context: Compactness Theorem is for Propositional Logic, and $\vDash$ here is tautological implication).

3. (3 points) Prove that if $\Sigma \vDash \beta$, then, there exists a finite set $\alpha_0, \cdots, \alpha_n$ such that $\alpha_0 \to \alpha_1 \to \cdots \to \alpha_n \to \beta$ is a tautology. (Here, $\vDash$ is tautological implication)

4. (4 points) Prove that $\Gamma \cup \{\alpha\} \vDash \beta$ if and only if $\Gamma \vDash \alpha \to \beta$. (Context: $\vDash$ here is Logical Implication)

5. (3 points) Prove that if $\vdash \alpha \to \beta$, then, $\vdash \forall x\alpha \to \forall x\beta$

6. (3 points) Prove that $\forall x\forall y[(x = y) \to (y = x)]$. (Context: Points were given if you proved validity or that this is a theorem)